

Appendix II. Definition of the analytic index of a relative elliptic complex

In this appendix we propose to show that the notion of a perfect complex leads to a simple and natural definition of the “analytic index” of a continuous family of elliptic differential operators, or more generally of a relative elliptic complex. For the initiated reader, let us say that the definition given here has the advantage, compared with the classical definitions (see for example [2] and [17]), of not requiring any deformation of the elliptic complex under consideration.

Sections 1 and 2 consist of reminders on the classical notion of a “mixed structure”.¹ We give, of course, only the minimum of definitions and results required in order to state and prove, in Section 3, the central theorem we have in view. It says, more precisely, that the direct image, by the projection (assumed proper) onto the base (assumed locally compact), of a relative elliptic complex is a perfect complex. When the base S is compact, the definition of the analytic index as an element of $K(S)$, the Grothendieck group of complex vector bundles of finite rank on S , then follows from the globalization sorite of II 2.3.2.

Throughout what follows, if S is a topological space, $\mathcal{O}_S$ will denote the sheaf of continuous real-valued functions on S .

1. The sorite of mixed structures

In this section a topological space S is fixed.

1.1. Definition of mixed varieties

Let V be an open subset of $\mathbb{R}^n$. For $r \in \mathbb{N}$ one writes $C^r(V)$ for the Fréchet space of real-valued functions of class C^r on V . Put

$$C^\infty(V) = \lim_r C^r(V).$$

Let U be an open subset of S . For $r \in \mathbb{N} \cup \{\infty\}$, denote by ${}^r\mathcal{O}_{U \times V}$ the sheaf of rings on $U \times V$ defined by

$$\Gamma(U' \times V', {}^r\mathcal{O}_{U \times V}) = C(U', C^r(V')),$$

where U' (respectively V') is an open subset of U (respectively of V), and $C(U', C^r(V'))$ denotes the space of continuous maps from U' into $C^r(V')$. There is a canonical injection

$$\mathrm{pr}_1^{-1}(\mathcal{O}_U) \longrightarrow {}^r\mathcal{O}_{U \times V}$$

which makes $(U \times V, {}^r\mathcal{O}_{U \times V})$ into a ringed space over U .

A ringed space X over S is said to be a *mixed variety of class C^r* ($r \in \mathbb{N} \cup \{\infty\}$) if X is locally isomorphic, as a ringed space over S , to a space of the type just described. If S is reduced to one point, a mixed variety of class C^r over S is just a variety of class C^r in the usual sense.

Let $f : X \rightarrow S$ be a mixed variety of class C^r . For $s \in S$, the fiber $X_s = f^{-1}(s)$, endowed with the sheaf $i_s^{-1}(\mathcal{O}_X)/I_s$, where $i_s : X_s \rightarrow X$ is the injection and I_s is the ideal of sections which vanish on X_s , is a variety of class C^r .

¹The informed reader is advised to skip them.

1.2. Relative differential operators

It is easy, on the model of algebraic geometry (EGA IV 16, SGA 3 VII), to develop an infinitesimal study of mixed varieties. We shall only review rapidly the notions we shall need.

1.2.1. Let $f : X \rightarrow S$ be a mixed variety of class C^∞ . For each $n \in \mathbb{N}$ one defines the sheaf $P_{X/S}^n$ of principal parts of order n on X . In two different ways, corresponding to the two projections pr_1 and pr_2 of $X \times_S X$ onto X , it is an $\mathcal{O}_X$ -algebra augmented toward $\mathcal{O}_X$, and is locally free of finite type as an $\mathcal{O}_X$ -module. When the $\mathcal{O}_X$ -module structure on $P_{X/S}^n$ is not specified, it will be understood to be the one coming from pr_1 . One has the canonical identifications

$$P_{X/S}^0 = \mathcal{O}_X, \quad P_{X/S}^1/\mathcal{O}_X = \Omega_{X/S}^1.$$

The sheaf $\Omega_{X/S}^1$ is the sheaf of relative differentials of X over S , or relative cotangent sheaf. Its dual is denoted $T_{X/S}$ and called the relative tangent sheaf. The sheaf $T_{X/S}$, respectively $\Omega_{X/S}^1$, is the sheaf of sections of a vector bundle of finite rank on X , called the relative tangent, respectively cotangent, bundle; these bundles are mixed varieties over X .

The $f^{-1}(\mathcal{O}_S)$ -linear homomorphism, corresponding to the $\mathcal{O}_X$ -algebra structure defined by pr_2 ,

$$d_{X/S}^n : \mathcal{O}_X \longrightarrow P_{X/S}^n,$$

which associates to a section of $\mathcal{O}_X$ its principal part of order n , is called the universal differential operator of order n on X relative to S .

1.2.2. More generally, if E is a locally free $\mathcal{O}_X$ -module of finite type, one may consider the sheaf of principal parts of order n of E ,

$$P_{X/S}^n(E) = P_{X/S}^n \otimes_{\mathcal{O}_X} E,$$

where the tensor product is defined by means of the second $\mathcal{O}_X$ -module structure on $P_{X/S}^n$. One has a universal differential operator

$$d_{X/S}^n(E) : E \longrightarrow P_{X/S}^n(E).$$

Let E, F be locally free $\mathcal{O}_X$ -modules of finite type. By definition, a differential operator from E to F of order $\leq n$ is an $f^{-1}(\mathcal{O}_S)$ -linear homomorphism from E to F which factors through $d_{X/S}^n(E)$ as an $\mathcal{O}_X$ -linear homomorphism $P_{X/S}^n(E) \rightarrow F$; the factorization is then unique. In other words, $d_{X/S}^n(E)$ gives a bijection

$$\text{Hom}_{\mathcal{O}_X}(P_{X/S}^n(E), F) \xrightarrow{\sim} \text{Diff}^n(E, F), \quad (1.2.2.1)$$

where $\text{Diff}^n(E, F)$ denotes the set of differential operators from E to F of order $\leq n$.

1.2.3. We shall also have to consider differential operators with complex coefficients. Let $\mathcal{O}_X \otimes \mathbb{C}$ (respectively $\mathcal{O}_S \otimes \mathbb{C}$) be the complexification of $\mathcal{O}_X$ (respectively $\mathcal{O}_S$). Let E, F be locally free $(\mathcal{O}_X \otimes \mathbb{C})$ -modules of finite type. By a differential operator from E to F of order $\leq n$ one means an $f^{-1}(\mathcal{O}_S \otimes \mathbb{C})$ -linear homomorphism from E to F which factors through $d_{X/S}^n(E)$ as an $(\mathcal{O}_X \otimes \mathbb{C})$ -linear homomorphism $P_{X/S}^n(E) \rightarrow F$. If $\text{Diff}_{\mathbb{C}}^n(E, F)$ denotes the set of such operators, $d_{X/S}^n(E)$ therefore defines a bijection

$$\text{Hom}_{\mathcal{O}_X \otimes \mathbb{C}}(P_{X/S}^n(E), F) \xrightarrow{\sim} \text{Diff}_{\mathbb{C}}^n(E, F). \quad (1.2.3.1)$$

1.2.4. For $n \geq 1$ there is a canonical exact sequence

$$0 \longrightarrow S^n(\Omega_{X/S}^1) \longrightarrow P_{X/S}^n \longrightarrow P_{X/S}^{n-1} \longrightarrow 0.$$

Hence, tensoring over $\mathcal{O}_X$ with a locally free $\mathcal{O}_X$ -module, respectively $(\mathcal{O}_X \otimes \mathbb{C})$ -module, E of finite type, one obtains an exact sequence

$$0 \longrightarrow S^n(\Omega_{X/S}^1) \otimes E \longrightarrow P_{X/S}^n(E) \longrightarrow P_{X/S}^{n-1}(E) \longrightarrow 0. \quad (1.2.4.1)$$

Here $S^n(\Omega_{X/S}^1)$ denotes the n -th symmetric power.

Let F be a locally free $\mathcal{O}_X$ -module, respectively $(\mathcal{O}_X \otimes \mathbb{C})$ -module, of finite type. Applying $\text{Hom}(_, F)$, respectively $\text{Hom}_{\mathcal{O}_X \otimes \mathbb{C}}(_, F)$, to (1.2.4.1), and using (1.2.2.1) and (1.2.3.1), one obtains an exact sequence

$$0 \longrightarrow \text{Diff}^{n-1}(E, F) \longrightarrow \text{Diff}^n(E, F) \xrightarrow{\sigma_n} \text{Hom}_{\mathcal{O}_X}(S^n(\Omega_{X/S}^1) \otimes E, F), \quad (1.2.4.2)$$

and analogously in the complex case. If X is paracompact and $\mathcal{O}_X$ is soft, the sequence may be completed on the right by adding a zero. The homomorphism σ_n is called the *symbol*. It associates to a differential operator from E to F of order $\leq n$ a homogeneous polynomial map of degree n from $T_{X/S}^*$ into $\text{Hom}(E, F)$, or equivalently a homomorphism $p^*E \rightarrow p^*F$, where p is the projection of the relative cotangent bundle onto X , satisfying the usual homogeneity condition.

Let E, F, G be locally free $\mathcal{O}_X$ -modules of finite type, $u \in \text{Diff}^n(E, F)$ and $v \in \text{Diff}^m(F, G)$. Then $vu \in \text{Diff}^{m+n}(E, G)$ and one checks at once the relation

$$\sigma_{m+n}(vu) = \sigma_m(v)\sigma_n(u). \quad (1.2.4.3)$$

The same relation holds for differential operators with complex coefficients. In particular, if the composite of two differential operators is zero, then the composite of their symbols is zero.

1.2.5. Let $E^\bullet$ be a bounded complex of differential operators with complex coefficients on X relative to S , and let $d^i : E^i \rightarrow E^{i+1}$ be of order n_i . By (1.2.4.3),

$$\sigma_{n_{i+1}}(d^{i+1})\sigma_{n_i}(d^i) = 0.$$

The complex $E^\bullet$ is called an *elliptic complex of order (n_i)* if the symbol complex on the relative cotangent bundle, pulled back to that bundle, is acyclic outside the zero section.

When S is reduced to one point, this notion of elliptic complex on X coincides with the usual one, studied in detail in [4] and [16].

1.2.6. Let $s \in S$, and let i_s be the inclusion of the fiber X_s in X , considered as a monomorphism of ringed spaces (1.1). For $n \in \mathbb{N}$ one has a canonical isomorphism

$$i_s^* P_{X/S}^n \simeq P_{X_s}^n,$$

and hence a restriction homomorphism

$$i_s^* : \text{Diff}^n(E, F) \longrightarrow \text{Diff}^n(i_s^*E, i_s^*F),$$

for E, F locally free $\mathcal{O}_X$ -modules of finite type, and similarly for operators with complex coefficients. If $E^\bullet$ is a bounded complex of differential operators with complex coefficients on X , then $E^\bullet$ is elliptic if and only if, for every $s \in S$, $i_s^*(E^\bullet)$ is an elliptic complex. The verification is left to the reader.

2. Rigidity lemmas

2.0. It is “well known” that compact differentiable varieties are rigid, that is, they admit no continuous deformation other than the trivial one. Since this result will be used essentially in

Section 3, and references seem rather scarce, it is perhaps useful to give precise statements and a few proofs.

A topological space S is fixed. The mixed varieties under consideration are assumed to be of class C^r , with $r \gg 1$; possibly $r = \infty$.

2.1. Notation. Let X, Y be mixed S -varieties. We denote by $\text{Hom}(X, Y)$ the set of S -morphisms from X to Y , and by $\mathcal{H}om(X, Y)$ the sheaf on X of germs of S -morphisms from X to Y , defined by

$$\mathcal{H}om(X, Y)(U) = \text{Hom}(U, Y)$$

for U open in X .

Let $s \in S$, and write i_s for the inclusion of X_s in X . There is a canonical induction homomorphism

$$i_s^{-1} \mathcal{H}om(X, Y) \longrightarrow \mathcal{H}om(X_s, Y_s) \quad (2.1.1)$$

which is an epimorphism, as is immediate. In the case $Y = S \times \mathbb{R}$, the sheaf $\mathcal{H}om(X, Y)$ is canonically identified with $\mathcal{O}_X$, and the epimorphism (2.1.1) with the epimorphism

$$i_s^{-1}(\mathcal{O}_X) \longrightarrow i_s^*(\mathcal{O}_X)$$

considered in (1.1).

Proposition 2.2. Let X, Y be mixed S -varieties, and let $s \in S$. Suppose that the fiber X_s is paracompact and that s has a fundamental system of paracompact neighborhoods. Let A be a closed subset of X_s , and let

$$f \in H^0(A, \mathcal{H}om(X_s, Y_s)).$$

The subsheaf $\mathcal{F}$ of $i_s^{-1} \mathcal{H}om(X, Y)$ consisting of the germs which extend f , that is, whose image by (2.1.1) coincides with f over A , is soft.

Proof. Let X' be an open neighborhood in X which is isomorphic, as a mixed variety, to $S' \times U$, where S' is a paracompact neighborhood of s and U is a relatively compact open subset of $\mathbb{R}^n$, and such that the image by f of $X'_s \cap A$ is contained in an open subset of Y isomorphic to $S' \times V$, where V is a convex open subset of $\mathbb{R}^q$ containing the origin. By T.F. II 3.4.1 it is enough to show that every section of $\mathcal{F}$ over a closed subset of X_s contained in X'_s extends to X'_s . Since $X'_s = \{s\} \times U$ is relatively compact, a closed subset of X_s contained in X'_s is compact. Identifying X' with $S' \times U$ and Y' with $S' \times V$, putting $A' = A \cap X'_s$, and using T.F. II 3.3.1, the assertion is reduced to the following elementary lemma.²

Lemma 2.2.1. Let U be an open subset of $\mathbb{R}^n$, and let V be a convex open subset of $\mathbb{R}^q$ containing 0. Let A_1, A_2 be closed subsets of $\{s\} \times U$. Let U_1 , respectively U_2 , be an open subset of $S \times U$, respectively of $\{s\} \times U$, containing A_1 , respectively A_2 . Let $f_1 : U_1 \rightarrow V$ be a morphism and let $f_2 : U_2 \rightarrow S \times V$ be an S -morphism which, on $U_1 \cap (\{s\} \times U)$, defines the same germ as f_1 near A_1 . Then there exists an open subset U' of $S \times U$ containing $\{s\} \times U$ and an S -morphism

$$F : U' \longrightarrow S \times V$$

such that the germ of F along $\{s\} \times U$ extends the germ defined by f_1 , and the morphism induced by F on $\{s\} \times U$ defines the same germ as f_2 near A_2 .

Proof. Replacing f_1 by af_1 , where a is a C^∞ function on U equal to 1 near A_1 and with support in U_1 , one may suppose $U_1 = U$. Let b be a C^∞ function on U , equal to 1 on A_2 and with support in $U_2 \cap (\{s\} \times U)$. Then

$$U' = U_1 \cup (S \times (U - \text{Supp } b))$$

²Communicated to the writer by A. Douady. A closed subset of a paracompact space has a fundamental system of paracompact neighborhoods.

is an open subset of $S \times U$ containing $\{s\} \times U$. The morphism $F : U' \rightarrow S \times V$ defined by

$$F(t, x) = (t, b(x) \operatorname{pr}_2 f_2(x) + (1 - b(x)) f_1(t, x))$$

visibly satisfies the required conditions.

Corollary 2.3. Under the hypotheses of (2.2), the sheaves $i_s^{-1} \mathcal{H}om(X, Y)$ and $\mathcal{H}om(X_s, Y_s)$ are soft. In particular, the sheaves $i_s^{-1}(\mathcal{O}_X)$ and $\mathcal{O}_{X_s}$ are soft. For every closed subset A of X_s , the map

$$H^0(X_s, i_s^{-1} \mathcal{H}om(X, Y)) \longrightarrow H^0(A, \mathcal{H}om(X_s, Y_s))$$

deduced from (2.1.1) is surjective.

Proof. The softness of $i_s^{-1} \mathcal{H}om(X, Y)$ follows from (2.2) applied with $A = \emptyset$. Taking $S = \{s\}$, one obtains the softness of $\mathcal{H}om(X_s, Y_s)$. The last assertion follows from (2.2).³

Corollary 2.4. Under the hypotheses of (2.2), let E be an $\mathcal{O}_X$ -module. For every closed subset A of X_s , the canonical homomorphism

$$H^0(X_s, i_s^{-1} E) \longrightarrow H^0(A, i_s^* E)$$

deduced from the epimorphism

$$i_s^{-1} E \longrightarrow i_s^* E = i_s^{-1} E \otimes_{i_s^{-1} \mathcal{O}_X} \mathcal{O}_{X_s}$$

is an epimorphism.

Proof. Since $i_s^{-1} \mathcal{O}_X$ is soft by (2.3), the functor $H^0(A, \)$ is exact on the category of $i_s^{-1} \mathcal{O}_X$ -modules (T.F. II 3.7.1, 3.4.2, 4.4.3). Hence

$$H^0(A, i_s^{-1} E) \longrightarrow H^0(A, i_s^* E)$$

is an epimorphism. Since $i_s^{-1} E$ is soft, the homomorphism

$$H^0(X_s, i_s^{-1} E) \longrightarrow H^0(A, i_s^{-1} E)$$

is an epimorphism, and this proves the assertion.

Corollary 2.5. Let $p : X \rightarrow S$ be a mixed variety and let $s \in S$. Suppose that s has a fundamental system of paracompact neighborhoods S' such that the $p^{-1}(S')$ form a fundamental system of paracompact neighborhoods of X_s . This condition is satisfied, for example, if S is locally compact and p is a proper map of topological spaces. Then, for every $\mathcal{O}_X$ -module E ,

$$(R^i p_* E)_s = 0 \quad (i > 0).$$

Proof. By T.F. II 4.17.1,

$$(R^i p_* E)_s = H^i(X_s, i_s^{-1} E).$$

But $i_s^{-1} \mathcal{O}_X$ is soft by (2.3), hence $H^i(X_s, i_s^{-1} E) = 0$ for $i > 0$.

Corollary 2.6. Let X, Y be mixed S -varieties, and let $s \in S$. Suppose that X satisfies the hypotheses of (2.5). For every morphism $f : X_s \rightarrow Y_s$, there exists an open neighborhood S' of s and a morphism

$$F : X|S' \longrightarrow Y|S'$$

such that $F_s = f$.

³For $\mathcal{O}_{X_s}$ this is very well known. The writer does not know whether $\mathcal{O}_X$ is soft when the hypotheses of (2.2) are satisfied for every $s \in S$ and X is paracompact. This would trivially imply, under those hypotheses, the softness of $i_s^{-1}(\mathcal{O}_X)$ as well as (2.5) below.

Proof. This is an immediate consequence of (2.3), with $A = X_s$, and of the fact that the hypotheses on X give

$$H^0(X_s, i_s^{-1} \mathcal{H}om(X, Y)) = \lim_{S' \ni s} \text{Hom}(X|S', Y|S')$$

for open S' containing s (T.F. II 8.5.5).

Proposition 2.7. Let X, Y be mixed S -varieties, let $f : X \rightarrow Y$ be a morphism, let $s \in S$, and let $x \in X_s$. If the tangent linear map of

$$f_s : X_s \longrightarrow Y_s$$

is invertible at x , then f is a local isomorphism at x ; that is, there is an open neighborhood U of x in X and an open subset V of Y such that f induces an isomorphism $U \simeq V$.

Proof. This is an easy consequence of the implicit function theorem. The details are left to the reader.

Corollary 2.8. Let X, Y be mixed S -varieties and let $s \in S$. Suppose that s has a fundamental system of neighborhoods S' such that the $X|S'$, respectively $Y|S'$, form a fundamental system of neighborhoods of X_s , respectively Y_s ; this is the case, for example, if S is locally compact and X , respectively Y , is proper over S . Let $f : X \rightarrow Y$ be an S -morphism such that

$$f_s : X_s \longrightarrow Y_s$$

is an isomorphism. Then there exists a neighborhood S' of s such that

$$f|X|S' : X|S' \longrightarrow Y|S'$$

is an isomorphism.

Proof. By (2.7), there exists an open neighborhood U of X_s in X such that $f|U$ is étale, that is, a local isomorphism. We may suppose that $U = X|S'$ for a neighborhood S' of s ; after shrinking S , we may therefore assume that f is étale. Then the fiber product $Z = X \times_Y X$ is naturally a mixed S -variety, étale over X for each projection, and the diagonal morphism $\Delta : X \rightarrow Z$ is an open immersion. Moreover the sets $Z|S'$ form a fundamental system of neighborhoods of Z_s . Since

$$\Delta_s : X_s \longrightarrow Z_s$$

is an isomorphism, it follows, after shrinking S , that Δ is an isomorphism. Thus f is an open immersion, and after shrinking S again it is an isomorphism.

Corollary 2.9. Let X be a mixed S -variety and let $s \in S$. Suppose that the hypothesis of (2.5) is satisfied. Then there exists a neighborhood S' of s in S such that $X|S'$ is S' -isomorphic to $S' \times X_s$.

Proof. Apply (2.6) and (2.8).

Corollary 2.10. Let $s \in S$ and let V be a paracompact variety. Put $X = S \times V$, and let i_s be the injection of V into X defined by s . Suppose that the hypothesis of (2.5) is satisfied. Let E be a locally free $\mathcal{O}_X$ -module, respectively $\mathcal{O}_X \otimes \mathbb{C}$ -module, of finite type. Then there exists a neighborhood S' of s in S such that $E|X|S'$ is isomorphic to $\text{pr}_2^*(i_s^*E)$, where $\text{pr}_2 : S' \times V \rightarrow V$ is the second projection.

Proof. Put $F = \text{pr}_2^*(i_s^*E)$. By (2.4), applied to $\mathcal{H}om(E, F)$, the induction homomorphism

$$\text{Hom}(i_s^{-1}E, i_s^{-1}F) \longrightarrow \text{Hom}(i_s^*E, i_s^*F)$$

is an epimorphism. On the other hand, by T.F. II 3.3.1,

$$\mathrm{Hom}(i_s^{-1}E, i_s^{-1}F) = \lim_{S' \ni s} \mathrm{Hom}(E|X|S', F|X|S').$$

Thus the canonical isomorphism $i_s^*E \simeq i_s^*F$ is induced, after shrinking S , by a homomorphism $f : E \rightarrow F$. Since f_s is an isomorphism, f is an isomorphism near X_s by Nakayama, hence over $X|S'$ for a suitable neighborhood S' of s .

3. The finiteness theorem

3.0. Conventions and notation

3.0.1. Let X and Y be topological spaces. We write $C(X, Y)$ for the set of continuous maps from X to Y , and $\mathcal{C}(X, Y)$ for the sheaf on X defined by $U \mapsto C(U, Y)$.

3.0.2. Let E, F be Banach spaces, real or complex. We write $L(E, F)$ for the Banach space of continuous linear maps from E to F .

3.0.3. The mixed varieties occurring in this section are assumed to be of class C^∞ . If X is a mixed S -variety, we shall often identify a locally free $\mathcal{O}_X$ -module, respectively $\mathcal{O}_X \otimes \mathbb{C}$ -module, of finite type with the corresponding real, respectively complex, vector bundle.

3.0.4. Let X be a compact ordinary variety, and let E be a real or complex vector bundle of finite rank on X . Write $C^\infty(X, E)$ for the Fréchet space of C^∞ sections of E on X . For $r \in \mathbb{Z}$, write $C_r(X, E)$ for the Hilbert space of sections of E which are r times differentiable in L^2 ([16] IX). For $s \leq r$, $C_r(X, E)$ maps to $C_s(X, E)$ by a continuous linear injective map with dense image, and

$$C^\infty(X, E) = \lim_r C_r(X, E).$$

3.0.5. Let S be a topological space. We shall need the notion of a Hilbert bundle over S , for which we refer the reader to [12] or [14]. There is an additive functor from the category of real, respectively complex, Hilbert bundles over S to the category of $\mathcal{O}_S$ -, respectively $\mathcal{O}_S \otimes \mathbb{C}$ -, modules: it sends a Hilbert bundle E to the sheaf of continuous sections of E .

3.1. Auxiliary technical results

Lemma 3.1.1. Let X, Y be topological spaces, with X locally compact. Let $\mathcal{C}'(X, Y)$ be the presheaf, on the basis of relatively compact open subsets of X , defined by

$$\mathcal{C}'(X, Y)(U) = C(\overline{U}, Y).$$

The canonical restriction morphism

$$\mathcal{C}'(X, Y) \longrightarrow \mathcal{C}(X, Y)$$

induces an isomorphism on the associated sheaves.

Proof. Immediate.

Lemma 3.1.2. Let E, F be locally convex topological vector spaces over $\mathbb{C}$, and let $u : E \rightarrow F$ be a continuous linear map, respectively a continuous linear map with dense image. Let X be compact. Endow $C(X, E)$ and $C(X, F)$ with the topology of uniform convergence. Then the induced linear map

$$u_X : C(X, E) \longrightarrow C(X, F)$$

is continuous, respectively continuous with dense image.

Proof. It is clear that u_X is continuous. On the other hand, by Bourbaki, *Intégration*, chapter II, §2, lemmas 1 and 2, the subspace $C(X, \mathbb{C}) \otimes E$ of $C(X, E)$, formed by the continuous maps from X to E whose values lie in a finite-dimensional vector subspace, is dense; and similarly $C(X, \mathbb{C}) \otimes F$ is dense in $C(X, F)$. If u has dense image, then $u_X(C(X, E))$ is dense in $C(X, \mathbb{C}) \otimes F$, and taking closures gives the assertion.

3.1.3. Let V be a compact variety, let E and F be complex vector bundles of finite rank on V , and let $n \in \mathbb{N}$. The space $\text{Diff}_{\mathbb{C}}^n(E, F)$, identified by (1.2.3.1) with

$$\text{Hom}(P^n(E), F) \simeq C^\infty(V, \text{Hom}(P^n(E), F)),$$

is naturally a Fréchet space. Let $u \in \text{Diff}_{\mathbb{C}}^n(E, F)$. One checks easily that the linear map $C^\infty(V, E) \rightarrow C^\infty(V, F)$ defined by u is continuous. If $L(C^\infty(V, E), C^\infty(V, F))$ denotes the space of continuous linear maps, one therefore has a linear map

$$\delta : \text{Diff}_{\mathbb{C}}^n(E, F) \longrightarrow L(C^\infty(V, E), C^\infty(V, F)).$$

Moreover, for $r \in \mathbb{Z}$, u acts by extension of scalars on the C_r sections of E and defines a linear map

$$\delta_r : \text{Diff}_{\mathbb{C}}^n(E, F) \longrightarrow L(C_r(V, E), C_{r-n}(V, F)).$$

The maps δ_r are continuous, and for $u \in \text{Diff}_{\mathbb{C}}^n(E, F)$, $\delta(u)$ is the inverse-limit element of the projective system of the $\delta_r(u)$. These assertions follow easily from [16] IX: one reduces immediately to $n = 0$; the statement on the C_r follows directly from the definitions, while the statement for C^∞ follows, after localizing by a partition of unity, from the formula for successive derivatives of a product. The analogous real statements are also valid.

Lemma 3.1.4. Let S be a topological space, let V be a compact variety, let E, F be complex vector bundles of finite rank on V , and let $n \in \mathbb{N}$. Put $X = S \times V$. For

$$u \in \text{Diff}_{\mathbb{C}}^n(\text{pr}_2^* E, \text{pr}_2^* F),$$

the map $s \mapsto i_s^* u$ from S into $\text{Diff}_{\mathbb{C}}^n(E, F)$ is continuous. The linear map

$$\text{Diff}_{\mathbb{C}}^n(\text{pr}_2^* E, \text{pr}_2^* F) \longrightarrow C(S, \text{Diff}_{\mathbb{C}}^n(E, F))$$

so defined is an isomorphism.

Proof. By (1.2.3.1) one may restrict to the case $n = 0$. Putting $G = \text{Hom}(E, F)$, one is reduced to showing that for $u \in \Gamma(X, \text{pr}_2^* G)$, the map $s \mapsto u_s = i_s^* u$ from S into $C^\infty(V, G)$ is continuous and that the resulting map

$$\Gamma(X, \text{pr}_2^* G) \longrightarrow C(S, C^\infty(V, G))$$

is an isomorphism. This follows immediately from the definition of mixed varieties.

3.1.5. Under the hypotheses of (3.1.4), let

$$u \in \text{Diff}_{\mathbb{C}}^n(\text{pr}_2^* E, \text{pr}_2^* F).$$

Identifying u , by (3.1.4), with a continuous map from S into $\text{Diff}^n(E, F)$, one sees that, for U open in S , the morphism

$$\text{pr}_{1*}(u)(U) : C^\infty(U, E) \longrightarrow C^\infty(U, F)$$

is canonically identified with the linear map

$$C(U, C^\infty(V, E)) \longrightarrow C(U, C^\infty(V, F))$$

deduced from

$$\delta u : U \longrightarrow L(C^\infty(V, E), C^\infty(V, F)).$$

Suppose S locally compact. By (3.1.1), the morphism of sheaves

$$\mathrm{pr}_{1*}(u) : \mathrm{pr}_{1*} \mathrm{pr}_2^* E \longrightarrow \mathrm{pr}_{1*} \mathrm{pr}_2^* F$$

is the morphism associated to the morphism of presheaves

$$\mathcal{C}'(S, C^\infty(V, E)) \longrightarrow \mathcal{C}'(S, C^\infty(V, F))$$

defined by δu . On the other hand, for $r \in \mathbb{Z}$,

$$\delta_r u : S \longrightarrow L(C_r(V, E), C_{r-n}(V, F))$$

defines a morphism of presheaves

$$u_r : \mathcal{C}'(S, C_r(V, E)) \longrightarrow \mathcal{C}'(S, C_{r-n}(V, F)).$$

By (3.1.3), the u_r form a projective system whose limit is the previous morphism. It will be convenient to interpret $\delta_r u$ as a morphism of trivial Hilbert bundles

$$S \times C_r(V, E) \longrightarrow S \times C_{r-n}(V, F),$$

inducing on sheaves of sections the sheaf morphism associated to u_r .

3.2. Definition of the index

Theorem 3.2.1. Let S be a locally compact space, let $p : X \rightarrow S$ be a proper mixed variety, and let $E^\bullet$ be an elliptic complex of order (n_i) of complex vector bundles on X (1.2.5). Then the complex $Rp_*(E^\bullet)$, which by (2.5) is isomorphic to $p_*(E^\bullet)$ in the derived category of $\mathcal{O}_S \otimes \mathbb{C}$ -modules, is perfect.

Proof. The question being local on S , one may suppose by (2.9) that $X = S \times V$, where V is a compact variety, and that $p = \mathrm{pr}_1$. Moreover, by (2.10), one may suppose that for all i ,

$$E^i = \mathrm{pr}_2^*(F^i),$$

where F^i is a complex vector bundle of finite rank on V . By (3.1.5), $p_*(E^\bullet)$ is associated to the complex of presheaves

$$\mathcal{C}'(S, C^\infty(V, F^\bullet)),$$

where the differentials are induced by the given differential operators. Put, for each $i \in \mathbb{Z}$,

$$m_i = \sum_{j < i} n_j,$$

with the convention $n_i = 0$ if $E^i = 0$. For each $r \in \mathbb{Z}$ one has, again by (3.1.5), a complex of presheaves

$$\mathcal{C}'(S, C_{r-m_i}(V, F^i)),$$

which we denote by $\mathcal{C}'(S, C_r(V, F^\bullet))$. These complexes form a projective system, and

$$\mathcal{C}'(S, C^\infty(V, F^\bullet)) = \lim_r \mathcal{C}'(S, C_r(V, F^\bullet)). \quad (3.2.1.1)$$

Finally, denote by $F_r^\bullet$ the complex of trivial Hilbert bundles over S defined by the corresponding $\delta_r(d^i)$, and by $\mathcal{F}_r^\bullet$ the complex of sheaves of sections; by (3.1.5), it is just the complex of

sheaves associated to $\mathcal{C}'(S, C_r(V, F^\bullet))$. To prove the theorem it is enough to prove the following assertions:

(a) For $r \in \mathbb{Z}$, the canonical morphism

$$\mathcal{C}'(S, C^\infty(V, F^\bullet)) \longrightarrow \mathcal{C}'(S, C_r(V, F^\bullet))$$

is a quasi-isomorphism of complexes of presheaves, that is, it induces isomorphisms on the presheaves of cohomology.

(b) For $r \in \mathbb{Z}$, the complex $\mathcal{F}_r^\bullet$ is perfect.

Proof of (b). The ellipticity of $E^\bullet$ implies ([16] XI, theorem 6) that, for every $s \in S$, the differential of $(F_r^\bullet)_s$ has closed image and the spaces $H^i((F_r^\bullet)_s)$ are finite-dimensional vector spaces. In the terminology of [11], $F_r^\bullet$ is a direct quasi-acyclic complex of Hilbert bundles. Hence, by [12] 2.3, $F_r^\bullet$ is locally homotopically equivalent to a bounded complex of vector bundles of finite rank. Since the sheaf-of-sections functor is additive (3.0.5), $\mathcal{F}_r^\bullet$ is locally homotopically equivalent to a bounded complex of locally free sheaves of finite type; hence it is perfect.

For the proof of (a) we shall need the following intermediate assertion:

(a') For $p \gg q$, the canonical morphism

$$F_p^\bullet \longrightarrow F_q^\bullet$$

is a quasi-isomorphism.

Proof of (a'). Let $M_{pq}^\bullet$ be the complex of Hilbert bundles which is the cone of the morphism $F_p^\bullet \rightarrow F_q^\bullet$. The ellipticity of $E^\bullet$ implies that, for every $s \in S$, the morphism $(F_p^\bullet)_s \rightarrow (F_q^\bullet)_s$ is a quasi-isomorphism ([16] XI, corollary 2 and theorem 6). In other words, $(M_{pq}^\bullet)_s$ is a direct acyclic complex. By [11] 2.3, $M_{pq}^\bullet$ is locally homotopically trivial; hence its sheaf complex, the cone of $\mathcal{F}_p^\bullet \rightarrow \mathcal{F}_q^\bullet$, is acyclic.

Proof of (a). It is enough to show that, for every relatively compact open subset U of S , the morphism

$$C^\infty(U, C^\infty(V, F^\bullet)) \longrightarrow C^\infty(U, C_r(V, F^\bullet))$$

is a quasi-isomorphism. Replacing S by $\bar{U}$ and $E^\bullet$ by the induced complex over $\bar{U}$, one is reduced to proving that

$$C(S, C^\infty(V, F^\bullet)) \longrightarrow C(S, C_r(V, F^\bullet))$$

is a quasi-isomorphism, with S compact. By (3.2.1.1),

$$C(S, C^\infty(V, F^\bullet)) = \lim_i C(S, C_i(V, F^\bullet)).$$

For each i , $C(S, C_i(V, F^j))$ is a Banach space for the topology of uniform convergence, and the transition morphisms

$$C(S, C_i(V, F^j)) \longrightarrow C(S, C_{i-1}(V, F^j))$$

are continuous linear maps with dense image by (3.0.4) and (3.1.2). On the other hand, applying to the quasi-isomorphisms (a') the functor “sections”, which is exact since $\mathcal{O}_S$ is soft, one sees that the transition morphisms are quasi-isomorphisms. Assertion (a) follows from EGA 0_I 13.2.3, topological variant. This completes the proof of (3.2.1).

3.2.2. If S is a topological space, we denote by $\text{Parf}(S)$ the triangulated category of perfect complexes of $(\mathcal{O}_S \otimes \mathbb{C})$ -modules, and by $K^0(S)$ the corresponding Grothendieck group.⁴

⁴See I 6.3. For a general review of K -functors, see Exposé IV.

Let $S, X, E^\bullet$ be as in (3.2.1). The *analytic index* of $E^\bullet$ is the element

$$\text{ind}_{\text{an}}(E^\bullet) = [Rp_*(E^\bullet)] \in K^0(S).$$

If S is compact, $\text{ind}_{\text{an}}(E^\bullet)$ canonically defines an element in the usual group $K(S)$ of topologists. Indeed, in general, for every topological space S , let $\text{Loclib}(S)$ be the additive category of locally free $(\mathcal{O}_S \otimes \mathbb{C})$ -modules of finite type, and let $K(S)$ be the Grothendieck group of the triangulated category

$$K^b(\text{Loclib}(S))/K_{\text{ac}}^b(\text{Loclib}(S)).$$

This category is in fact identified with $K^b(\text{Loclib}(S))$ when S is paracompact (II 2.2.2(a)), so in this case, by I 6.4, $K(S)$ is canonically isomorphic to the Grothendieck group of the additive category $\text{Loclib}(S)$. There is a natural homomorphism

$$K(S) \longrightarrow K^0(S).$$

By II 2.3.2 this is an isomorphism for compact S , proving the assertion. More generally, $\text{ind}_{\text{an}}(E^\bullet)$ defines an element in the completed group

$$\widehat{K}(S) = \lim_A K(A),$$

the projective limit being taken over the compact subsets A of S .

3.2.3. The writer does not know whether the finiteness theorem (3.2.1) remains valid if, instead of assuming S locally compact, one merely assumes that each point of S has a fundamental system of paracompact neighborhoods. Moreover, the local compactness hypothesis on S does not seem very natural. The space of “universal local parameters” for elliptic complexes of fixed order is in fact a subspace of infinite dimension in a Fréchet space. More precisely, let V be a compact variety, let $F = (F^i)$ be a sequence of complex vector bundles of finite rank on V , zero except for finitely many indices, and let $n = (n_i)$ be a sequence of integers ≥ 0 . Let $\text{Comp. diff}_n(F)$ be the space of complexes of differential operators of base F and order n , that is, the closed subspace of

$$\prod_i \text{Diff}^{n_i}(F^i, F^{i+1})$$

defined by the equations $d^{i+1}d^i = 0$. This product is a Fréchet space. The subset $\text{Ell}_n(F)$ formed by the elliptic complexes of order n is open. It is this space, in general not locally compact, which plays the role of a space of universal local parameters.

3.2.4. Nevertheless, even if the generalization of (3.2.1) just mentioned is impossible, one can always associate to $E^\bullet$, under the sole hypothesis that the fibration p is proper and locally trivial, an element of $K^0(S)$ deserving the name analytic index of $E^\bullet$. Indeed, suppose, for simplicity, that we are in the local situation where $X = S \times V$ with V compact, and $E^\bullet$ is given by a continuous map from S into $\text{Ell}_n(F)$ (3.2.3). Assertions (a') and (b) in the proof of (3.2.1) are in fact valid without any hypothesis on S ; it is therefore reasonable to put, with the notation of that proof,

$$\text{ind}_{\text{an}}(E^\bullet) = [\mathcal{F}_r^\bullet] \in K^0(S),$$

this element being independent of r .

3.2.5. Suppose S paracompact. Let H be a complex Hilbert space of infinite Hilbert dimension, and let $I(H)$ be the subspace of $L(H, H)$ formed by the Fredholm operators. The complexes $F_r^\bullet$ define [12] a canonical element in the group $[S, I(H)]$ of homotopy classes of continuous maps from S to $I(H)$. But the writer does not know, for non-compact S (even when S is locally compact), what the relations are between $[S, I(H)]$ and $K^0(S)$: is there a natural arrow from

one to the other? are they isomorphic? A fortiori, the relation between the two indices thus defined remains unclear.

For a generalization, under the hypotheses of (3.2.1), of the Atiyah–Singer index theorem, we refer to the recent work of Shih [17], or to work in preparation of M. F. Atiyah and I. M. Singer.⁵

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⁵For the general case, use the notion of “weak Hilbert bundle”, or continuous field of Hilbert spaces, of [12].